Phase 16.5: Personal Growth Context Injection & Naturalness Gate

PHASE 16.5 FROZEN

Architecture Freeze Record, Ground Truth Invariants & Golden Verification Matrix

Date: 2026-10-04	**Architecture Status:** Gate 0 Approved (Rev 3 + 4 Amendments) 🔒 FROZEN	**Target Milestones:** 16.1 🔒, 16.2 🔒, 16.3 🔒, 16.4 🔒, 16.5 🔒
Total Backend Tests: 2,568 / 2,568 (100% PASS)	**Phase 16 Master Tests:** 118 / 118 (100% PASS, G01 ~ G80)	**Desktop Build:** 0 Errors (Vite + TypeScript Clean)

1. Executive Summary & Architectural Scope

Phase 16.5 accomplishes the crucial integration step of Project Aevum: bridging the canonical ground-truth growth reality (experiences, facet appraisals, longitudinal satiation/dormancy) into the companion's living conversational self, while dismantling the "interrogation bot / questionnaire assistant" syndrome.

> **Core Axiom:** "*Memory 是材料，不是台词*" (Memory is cognitive substance, not recited scripts). Context injection informs the model who Aevum is, what she has lived, what she knows objectively, and her relational distance, while conversational policy ensures authentic curiosity, dynamic length, and natural statement closures.

2. Core Architectural Pillars & Final Amendments

Pillar 1: Two-Dimensional Epistemic Matrix & Provenance Hard Invariants (Amendment 1)

Decoupled experience depth (`ExperienceStatus`) from information provenance (`EpistemicProvenanceKind`):

- `UNFAMILIAR / HEARD_OF / EXPERIENCED` × `WORLD_KNOWLEDGE / SELF_EXPERIENCE / SELF_APPRAISAL / USER_REPORTED / INFERRED`.
- **Hard Invariant:** `USER_REPORTED` is isolated to user context and never mutates Aevum's own taste or preferences.
- **Hard Invariant:** `INFERRED` deductions are never silently promoted to lived experiences or canonical preferences.
- **Epistemic Humility:** Topics with `HEARD_OF` provenance admit lack of personal exposure; encyclopedic Wikipedia summaries are forbidden.

Pillar 2: Versioned Genesis Blueprint & Relationship Contracts (Amendment 2)

- **Tier A (`CharacterBlueprintConfig`):** Versioned setup blueprint. Bumping versions preserves continuity and never deletes or truncates historical lived experiences.
- **Tier B (`RelationshipGenesisConfig`):** Discrete baseline parameters mapped *strictly once* during genesis. Never acts as a continuous runtime override over evolved relationship state.
- **Tier C (Autonomous Runtime Evolution):** Canonical runtime models (`RelationshipState`) evolve freely through real interactions without interference from UI genesis sliders.

Pillar 3: Growth Context Adapter Integration & Budget Precedence (Amendment 4)

- **Strictly Read-Only:** `PersonalGrowthContextAdapter` performs 0 DB writes, 0 experience creations, and 0 preference mutations. Verified by before/after database state hashes.
- **Budget Precedence Hierarchy:** Context budget (~1,200 chars soft target) prioritizes hard invariants (epistemic separation, current activity state, active domains) over low-relevance preferences. Critical boundaries are never truncated.
- **Determinism:** Context XML blocks are cryptographically verified via deterministic SHA-256 integrity digests.

Pillar 4: Conversational Naturalness & Anti-Questionnaire Policy (Amendment 3)

- **Memory as Cognitive Substance:** Prohibits "proof-of-retrieval" recitations (e.g., "*I remember you told me on Tuesday that...*"). Memories inform emotional resonance and subtext naturally.
- **Anti-Questionnaire Policy:** Questions arise only when genuinely curious. Turn closures are free to be simple statements, observations, or quiet pauses.
- **Anti-Parrot Policy:** Prohibits compulsive summarization openings (e.g., "*You like anime, which is great...*"). Responses begin with spontaneous reactions or direct thoughts.
- **Dynamic Response Length:** Length freely adapts from concise phrases (1 sentence) to thoughtful commentary (3-5 sentences) based on emotional moment, free of rigid length templates.

3. Verification & Golden Test Matrix (Phase 16.5: G66 ~ G80)

Case	Invariant / Target Tested	Verification Scenario	Result
G66	2D Epistemic Orthogonality	ExperienceStatus decoupled from EpistemicProvenanceKind; verify item classifications	**PASS**
G67	INFERRED Non-Promotion	Inferred provenance rejected from claiming lived experience status without grounded evidence	**PASS**
G68	USER_REPORTED Isolation	User stating preference does not create Aevum self-preference or self-experience	**PASS**
G69	Epistemic Humility XML	HEARD_OF produces explicit humility directive; forbids encyclopedic lecture output	**PASS**
G70	Blueprint Versioning	Blueprint version incrementation updates demeanor without mutating historical experiences	**PASS**
G71	Relationship Discrete Mapping	Initial parameters map discretely to runtime baseline without composite score collapse	**PASS**
G72	Genesis Non-Override Invariant	Genesis parameters never override 6-month evolved runtime relationship state	**PASS**
G73	Adapter SHA-256 Determinism	Identical state produces identical XML with matching cryptographic SHA-256 digest	**PASS**
G74	Adapter Read-Only Guarantee	Snapshot before == snapshot after across all Phase 16 tables after context assembly	**PASS**
G75	Budget Precedence Hierarchy	Tight character budget truncates low-relevance preferences while preserving critical invariants	**PASS**
G76	Anti-Parrot & Anti-Interrogation	Romance/anime discussion avoids formulaic restatements and compulsive question closures	**PASS**
G77	Epistemic Humility Fixture	Direct probe on unplayed game triggers authentic admission of non-play rather than fake critique	**PASS**
G78	Statement Closure Freedom	Turn closure without interrogative ending verified as valid and natural conversational stance	**PASS**
G79	Memory Substance Integration	Recalled memory woven into thoughts without artificial "retrieval proof" announcement	**PASS**
G80	Dynamic Response Length	Pacing adjusts dynamically to emotional weight; no rigid word/sentence bounds violated	**PASS**

4. Complete Phase 16 Golden Test Status

Sub-Phase	Scope	Tests	Constitutional Locks	Status
Phase 16.1	Experiential Growth Contracts & Integrity	17 / 17	Lock 1 (GC-401) ~ Lock 4 (GC-404)	**FROZEN**
Phase 16.2	State Machine & Activity Transitions	13 / 13	Lock 5 (GC-405) ~ Lock 6 (GC-407)	**FROZEN**
Phase 16.3	Facet-Level Subjective Appraisal	49 / 49	Lock 7 (GC-408) ~ Lock 9 (GC-412)	**FROZEN**
Phase 16.4	Longitudinal Dynamics (Fatigue, Dormancy, Resurgence)	24 / 24	Lock 10 (GC-413) ~ Lock 14 (GC-417)	**FROZEN**
Phase 16.5	Growth Context Injection & Naturalness Gate	15 / 15	Amendments 1, 2, 3, 4 (G66 ~ G80)	**FROZEN**
Phase 16 Complete Master Suite		**118 / 118**	**100% PASSING (1.85s)**	

5. Sign-Off & Phase Transition

All architectural constraints, constitutional locks, and naturalness requirements for Phase 16.5 are fully satisfied and verified. Phase 16 (Character Personal Growth & Naturalness) is hereby officially completed and **FROZEN**.